T2D Glucose Digital Twin

Predictive Metabolic Monitoring via Physics-Residual Temporal Fusion Transformers

Happiest Health Digital Twin Challenge — Final Submission

Target Use Case: Early Warning & Continuous Glucose Forecasting for Prediabetes and Type 2 Diabetes.

Team: T2D Digital Twin Team (Akshaya R — BGS College of Engineering and Technology; Nikhil VS — East West Institute of Technology)

Target Platform: Web-based Clinical Workstation & Replay Dashboard.

Release branch: release/candidate-clean

Investigational Research Prototype Notice: This software is an engineering research prototype designed to explore physiological modeling and deep learning integration on synthetic cohorts. It has not been clinically evaluated, certified as a medical device, or tested for autonomous therapeutic decisions.

RESEARCH PROTOTYPE

From a Reactive Reading to a Predictive Trajectory

The Clinical Problem and Monitoring Hypothesis

The Current Standard: Reactive Point Alarms

- Traditional Continuous Glucose Monitors (CGMs) alert after glucose breaches static thresholds (e.g., >180 mg/dL).
- Post-prandial glucose excursions occur rapidly (30–60 min rise time); threshold alarms leave minimal intervention window.
- Lack of physiological context leads to alert fatigue and high false alarm rates.

Our Approach: Continuous Forward Visibility

- Forecast future glucose trajectories across a **120-minute horizon** in 5-minute discretizations.
- Quantify predictive uncertainty via **quantile intervals** (p05 to p95; the P10–P90 interval is conformal-calibrated, the others are not separately evaluated).
- Estimate cumulative spike hazard and adverse event probability in real-time.
- Decompose contributing factors into mechanistic and behavioral drivers.

Clinical Boundary: The clinical hypothesis is that forward visibility aids timely dietary review; measured patient health outcomes or HbA1c reductions have not yet been evaluated in clinical trials.

Two Information Streams, One Patient Context

Observable Static EHR and Dynamic Telemetry Ingestion

Stream 1: Static EHR Patient Context

Captured at baseline / onboarding via validated schema:

- Demographics: Age, sex, BMI, weight.
- Metabolic Markers: Baseline HbA1c, eGFR, basal fasting glucose.
- Clinical History: Diabetes duration, diagnostic status (Healthy, Prediabetes, T2D).
- Medication Context: Metformin, SGLT2i, GLP-1 RA, DPP-4i, insulin.

Stream 2: Dynamic Telemetry & Wearable Streams

Ingested continuously on a synchronized 5-minute temporal grid:

- Continuous Glucose Monitoring (CGM) readings (mg/dL).
- Cardiorespiratory Wearable Signals: Heart Rate (HR), Heart Rate Variability (HRV).
- Physical Activity & Sleep: Step counts, sleep stage indicators.
- Logged Dietary Events: Carbohydrate intake timestamps and meal size.

Strict Observable-Input Boundary (Contract S3)

The deployed model strictly consumes observable clinical and wearable features. Privileged simulation latents (unannounced meal sizes, true plasma insulin) are isolated to the simulator environment and prohibited from entering model inputs.

VERIFIED INPUT CONTRACT

Hybrid Architecture: Physics Prior + Deep Residuals

Unscented Kalman Filter S3 + Temporal Fusion Transformer

1. Mechanistic UKF Prior

Two-compartment Bergman minimal glucose-insulin model with gut absorption dynamics:

- Tracks unobserved physiological states (insulin sensitivity, carbs-on-board).
- Provides deterministic physiological rollouts over 120 minutes.

2. PR-TFT Neural Architecture

Physics-Residual Temporal Fusion Transformer:

- Variable Selection Networks (VSN) for static and time-varying inputs.
- LSTM encoder/decoder capturing temporal dependencies.
- Causal multi-head self-attention for long-range interactions.

3. Residual Combination

The neural network learns the *residual error* over the physics prior:

- Final forecast = UKF physiological prior + PR-TFT learned quantile residuals.
- Predicts 7 quantiles (p05, p10, p25, p50, p75, p90, p95).
- Separate hazard head predicts spike probability.

Architectural Finding: Arm D (PR-TFT + UKF S3) anchors the forecast to a physiology twin and lets the network learn a residual; the residual can still add isolated one-step jumps (19.8% of sampled validation forecasts), which the product flags.

What the Model Returns: Trajectories & Risk

Standardized ForecastResponse v1 API Schema

Multi-Horizon Quantile Trajectory

- **Temporal Horizon:** 120 minutes forward from current as_of timestamp.
- **Discretization:** 24 discrete 5-minute timesteps.
- **Probabilistic Quantiles:** p05, p10, p25, p50 (median), p75, p90, p95.
- **Conformal Calibration:** Saved conformal interval widening fits the empirical 80% (p10–p90) confidence band with isotonic hazard calibration.

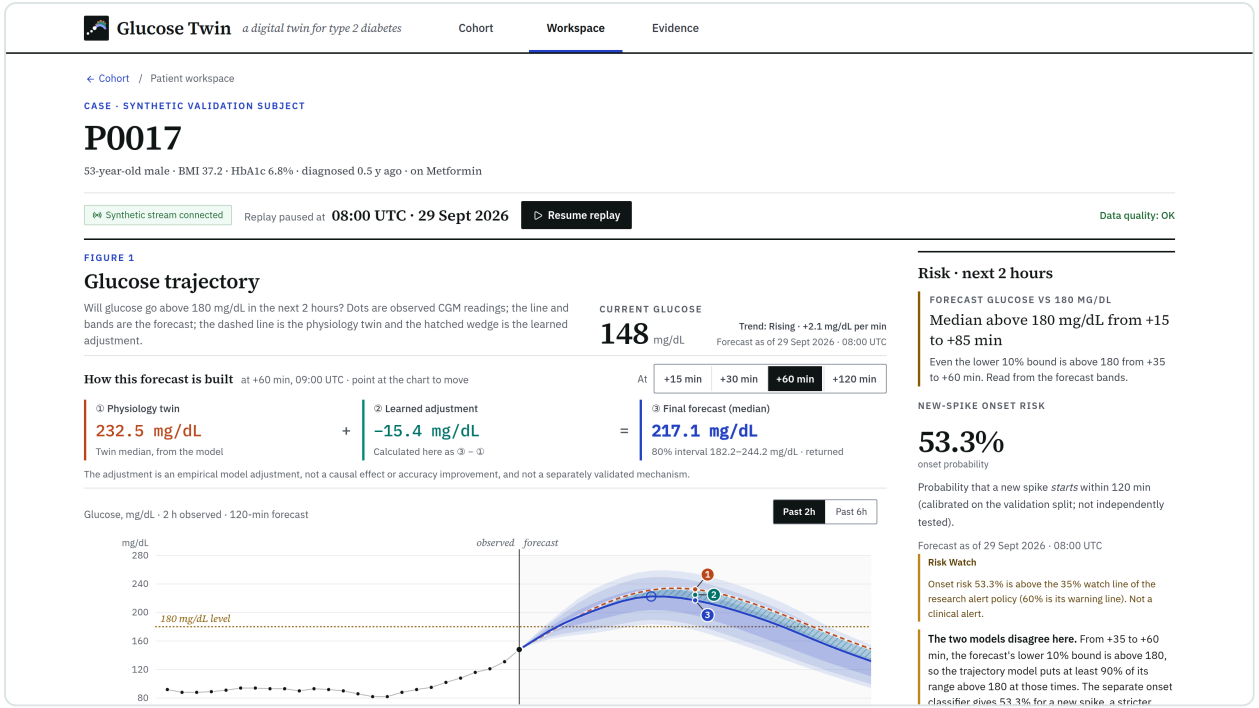
Adverse Event Risk Estimation

- **Spike Event Definition:** Glucose rising ≥ 50 mg/dL within 60 min and exceeding 180 mg/dL for ≥ 15 min.
- **Cumulative Risk Curve:** Onset probability evaluated across 5-min to 120-min offsets.
- **Risk Tiers:** Low (<35%), Watch (35–60%), High/Warning (≥60%).
- **Stateful Alert Policy:** Hysteresis filtering and 120-min suppression prevent alert flapping.

Verified Contract: All output packets strictly validate against Zod and Pydantic ForecastResponse schemas before reaching client presentation layers.

The Clinician Workstation (Live Screen)

Validation subject P0017 · EHR record + CGM and wearable streams → 2-hour forecast



One record, one time axis

EHR summary in the case header; the replay holds at 08:00 with CGM dots, the meal log, steps, heart rate, HRV and sleep under the same axis.

A forecast you can take apart

① physiology twin (dashed) + ② learned adjustment (hatched wedge) = ③ forecast (line, with 50/80/90% bands), on the chart and in the equation above it.

Risk, stated honestly

When the median is above 180 mg/dL, and a separate calibrated onset risk; when the two disagree, the product says so.

Captured from the running prototype (synthetic data, development-validation subject). Not a clinical device.

LIVE UI CAPTURE

Adverse Event Warning: Beyond Raw AUROC

Balancing Sensitivity, False Alarm Rates, and Decision Windows

Clinical Evaluation Dimensions

- **Event Sensitivity:** Proportion of actual spike episodes detected in advance.
- **False Alerts per Patient-Day (FA/day):** Ensuring alert thresholds do not overwhelm clinical workflows.
- **Effective Lead Time:** Median advance warning window (minutes before glucose breaches threshold).
- **Decision Actionability:** Whether the warning provides sufficient time for dietary adjustment.

Evaluator vs. Serving Policy Disclosures

- **Serving Policy:** Employs stateful hysteresis (Watch/Warning) and 120-min suppression for UI display.
- **Scientific Evaluator:** Uses independent, stateless point-in-time scoring rules for benchmark comparisons.
- **Warning Lead Time Limit:** A 120-min prediction horizon does *not* imply a 120-min warning lead time. Actual lead times depend on event velocity.

Audit Disclosure: The formal EW-01 scientific evaluation harness remains unexecuted due to incomplete governance sign-offs; no speculative superiority claims over clinical baselines are made.

Explaining the Model Without Claiming Causality

Arithmetic Driver Decomposition in the RiskPanel

Arithmetic Driver Decomposition

The RiskPanel visualizes the relative impact of model features on the predicted glucose trajectory:

- **Learned Correction:** Deep residual network delta over the physics prior.
- **Expected Habitual Meal:** Population-level meal timing probability.
- **Logged Meals Absorbing:** Mechanistic gut absorption from patient logs.
- **Return to Baseline:** Physiological clearance and insulin action.

Critical Interpretability Guardrails

- **Internal Decomposition:** Bars represent the mathematical contribution of components to the model's net output (mg/dL impact).
- **Not Causal Biology:** Feature weights do not establish direct biological causality in a living patient.
- **Unannounced Meal Detection:** Disclosed as a heuristic based on UKF innovation residuals, not verified ground truth.

UI Notice: The dashboard includes an explicit disclaimer: *“Arithmetic of the model’s own forecast — not a causal attribution.”* The contributions add up to the highest point of the forecast median.

What-If Simulation: Capability & Boundary

Interactive, backend-driven scenarios in the patient workspace (live screen)

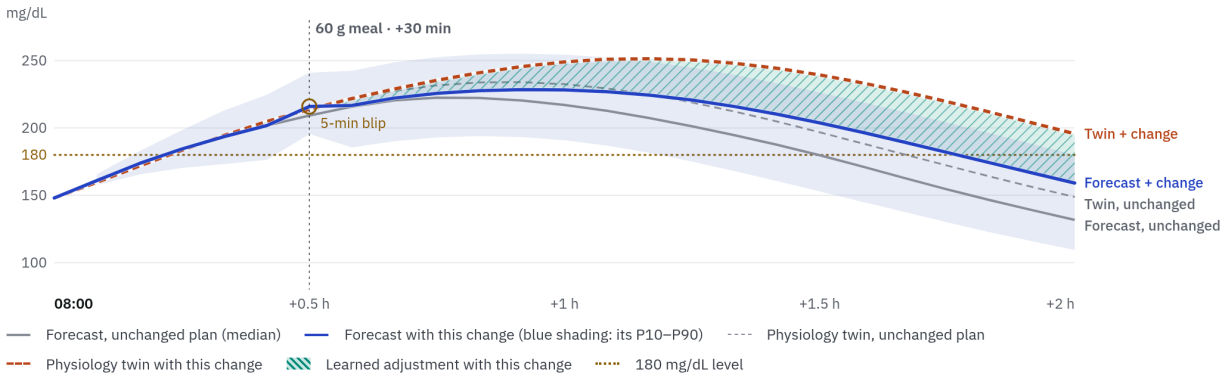
Result: 60 g carbohydrate meal in 30 min · from 08:00 UTC

Scenario inputs outside the training range. The network was not trained on inputs like this, so compare its curve with the physiology twin's.

▶ Why

Figure 2. Glucose under each plan

Next 2 h from 08:00 UTC. Grey: unchanged plan (median). Colour: with this change, with its 80% band.



Real backend response

The twin re-rolls the plan and the network re-scores it from the same anchor. The panel shows both curves, the twin-alone change and the calibrated risk.

Boundaries shown on screen

- “Simulated model response — not a validated causal treatment effect.”
- Meal inputs outside the network’s training range are flagged by the service.
- Medication changes are refused (SCENARIO_UNSUPPORTED).

P0017, 08:00 UTC, a further 60 g meal in 30 min: the physiology twin alone adds up to +47.8 mg/dL; the network’s highest median moves from 222.5 to 228.5 mg/dL. In 19.8% of sampled forecasts the learned adjustment adds an isolated 5-min step, which the chart flags (docs/evidence/model_audit). Values: docs/evidence/whatif_probe_P0017.json.

BOUNDARY DISCLOSURES

The Scientific Evidence Ledger

Separating Verified Implementation, Historical Metrics, and Unexecuted Work

Tier 2: Matched Validation Benchmarks

44 PATIENTS (54,281 ELIGIBLE ANCHORS / 53,849–53,920 EVALUABLE)

- **Matched RMSE (mg/dL · 95% Patient-Cluster CI, B=10,000 draws):**
  - **30m:** PR-TFT **12.36** [11.89–12.84] vs. UKF 15.24 [14.30–16.16] vs. Persist 21.64 [20.44–22.79]
    Δ -9.28 vs Persist [-10.05, -8.43], -2.88 vs Twin [-3.37, -2.38]; Rel Red 42.87% / 18.90% (10k draws < 0)
  - **60m:** PR-TFT **19.53** [18.56–20.48] vs. UKF 23.23 [21.76–24.64] vs. Persist 36.05 [33.85–38.13]
    Δ -16.53 vs Persist [-17.77, -15.17], -3.71 vs Twin [-4.27, -3.15]; Rel Red 45.84% / 15.96% (10k draws < 0)
  - **120m:** PR-TFT **25.03** [23.67–26.27] vs. UKF 28.64 [26.67–30.51] vs. Persist 48.36 [45.06–51.46]
    Δ -23.33 vs Persist [-25.31, -21.23], -3.61 vs Twin [-4.44, -2.81]; Rel Red 48.24% / 12.61% (10k draws < 0)
- **Patient Wins (Exact 95% Binomial CI):** 44/44 (100% [92.0–100%]) at 30m & 60m; 120m: 44/44 (100%) vs Persist, 41/44 (93.2% [81.3–98.6%]) vs Twin. *Cohort-limited Bernoulli assumption.*
- **Subgroup Robustness (RMSE60):** Healthy 19.31 (n=15, 18.5k elig / 18.4k eval), Prediabetes 19.87 (n=19, 23.5k elig / 23.3k eval), Type 2 19.18 (n=10, 12.3k elig / 12.2k eval).
- **Resubstitution Diagnostics:** 80% conformal coverage 80.00% (width 37.3 mg/dL); Brier score 0.0703 (raw 0.1650 on 47.6k at-risk anchors). Evaluated on val split; not clinical operating recommendations.
- *Boundary:* Validation cohort used in model selection; reflects simulated development validation, not human generalization.

Tier 1: Verified Implementation

CONFIRMED BY CODE & TESTS

- Observable-input contract enforcement.
- PR-TFT + UKF S3 model architecture.
- Chunk-invariant SSE proxy & recovery hook.
- 48 Jest suites (450 tests) and 162 backend tests (with RUN_DIR set) pass on the release branch.

Tier 3: Unexecuted Work

EXCLUDED FROM CLAIMS

- EW-01 scientific evaluation harness remains unexecuted.
- Demo baseline references missing checkpoint; metrics excluded.
- Zero prospective clinical outcome claims.

Evidence Integrity: We do not manufacture benchmark bars or promote historical training-set validation metrics as prospective generalization proof.

Reproducibility & Scientific Limitations

Synthetic Data Realism Gate 3 and Governance Status

Synthetic Data Realism Audit (Gate 3)

Gate 3 Result: FAILED
Discriminator AUROC: **0.9320** (historical, 90% CI: 0.928–0.936) / **0.9290** (duration-corrected, 90% CI: 0.925–0.933) vs. Target: **≤ 0.65**.

- A gradient-boosted discriminator reliably distinguishes synthetic ODE trajectories from mock-real generator records (top tell: jerk_sd, SHAP 0.8878; roc_max_fall).
- Factorial experiments reported in the validation report (scripts not in this repository) indicate AR(1) drift (SD 8–16 mg/dL) accounts for partial curvature divergence, but multivariate ODE dynamics remain distinguishable. Null control: AUROC 0.507.
- **Hazard Calibration (Resubstitution Diagnostic):** Empirical resubstitution diagnostics on 47,629 at-risk anchors show 14.2% anchor alert rate (sensitivity 63.0%, precision 58.7%) at threshold 0.35, and 5.3% anchor alert rate (sensitivity 29.9%, precision 74.5%) at threshold 0.60. Evaluated thresholds are empirical diagnostics on the validation set, not approved clinical operating points; anchor alert fractions do not represent event-level alerts per day.

Governance & Environmental Constraints

- **EW-01 Evaluation Unexecuted:** Protocol amendments remain pending co-author sign-offs; no experimental outcomes are reported.
- **Missing Baseline Checkpoint:** pr_tft_demo references an absent checkpoint; its baseline numbers are strictly excluded.
- **Clean Environment Verified:** Node 22, Webpack dev server, Python 3.12 venv, and isolated Git worktree verified.

Honest Science: Full scientific transparency requires reporting negative validation gates rather than concealing simulation limitations.

Demonstrated Scope & Research Roadmap

What is Complete, What is Frozen, and What Lies Ahead

Demonstrated & Frozen in this Release

- **Release branch:** release/candidate-clean; 48 Jest suites (450 tests) and 162 backend tests (with RUN_DIR set) pass.
- **End-to-End Pipeline:** Live synthetic backend → Next.js SSE proxy → browser patient dashboard verified.
- **Framing Resilience:** Chunk-invariant earliest-delimiter parser verified across all split boundaries.
- **Strict Validation:** Comprehensive fail-closed error handling and contract enforcement.

Future Clinical Research Horizons

- **Real Sensor Ingestion:** Hardware connectivity with Dexcom, Abbott FreeStyle Libre, and wearable APIs.
- **Prospective Clinical Trials:** Multi-center observational studies evaluating actual glycemic variability and HbA1c impact.
- **Causal Counterfactuals:** Clinically validated pharmacodynamic and lifestyle intervention models.
- **Medication What-Ifs:** Meal and activity What-Ifs ship now as simulated model responses; medication changes stay refused until the twin has validated drug terms.

Release Status: Code, documents and demo script are reconciled on release/candidate-clean. The demonstration video is pending recording.